INCH-POUND

MIL-W-71069 (AR)
19 March 1992

MILITARY SPECIFICATION

WARHEAD, 2.75 INCH ROCKET, FLECHETTE: XM255E1

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for the XM255E1 Flechette Warhead, a component of the Hydra 70 Rocket System. It covers the quality assurance provisions and special test requirements not covered by the drawings for parts and assembly of the flechette warhead (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement hereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-F-48877	-	Fuze, Rocket, M439 Parts and Assembly with Detonator

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-331	-	Fuzes and Components, Environmental Testing and Performance of
MIL-STD-810	-	Environmental Test Methods
MIL-STD-1167	-	Ammunition Data Cards
MIL-STD-1168	-	Ammunition Lot Numbering

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.3)

US ARMY ARMAMENT, RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

8796521	-	Marking and Sealing for Ammunition Fiber Containers
8796522	-	Marking Diagram and Sealing for Wood Packing Boxes
9230116	-	Packing and Marking of Wood Box for 2.75 Inch Rocket with Warhead, M229, M243, XM245, XM256, XM259, M261, M267, XM255E1 or ATAF
9335610	-	Packing and Marking for Container Ammunition, Fiber: PA89 for Rocket 2.75 Inch: Warhead M261 or M267
13058370-13	-	Rocket, 2.75 Inch, Cargo Warhead w/MK66 Motor
13282664-1	-	Warhead, 2.75 Inch Flechette: XM255E1
233AS431	-	Rocket Motor, Inert, MK66

OTHER PUBLICATIONS

1-2-601	-	Laboratory Inspection Schedules, International Test Operations Procedures (ITOP), TECOM

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(Copies of other Government documents, drawings, and publications required by Contractors in connection with specified acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials, processes and parts.

3.1.1 Components and assemblies. The components and assemblies shall comply with all requirements specified on drawing 13282664-1 and associated drawings with all requirements specified in applicable specifications and standards.

3.1.2 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.1.3 M439 Fuze. The fuze shall comply with Specification MIL-F-48877.

3.1.4 Functional straightness. Each warhead when attached to an inert rocket motor shall be capable of passing through a gage 66.178 + .250 inches long, 2.7952 + .0020 inches in diameter, and straight within .0020 inches total.

3.2 Physical characteristics.

3.2.1 Warhead weight. The weight shall not exceed 14.0 pounds.

3.3 Test requirements.

3.3.1 Temperature. The warhead shall withstand exposure to and function after a storage temperature of -50°F to 150°F.

3.3.2 Leakage. The warhead shall withstand exposure to the leakage test as specified in 4.5.6.

3.3.3 Helicopter vibration. The warhead shall withstand exposure to helicopter vibration as specified in 4.5.7 and shall comply with the ballistic test as specified in 4.5.2.

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3.3.4 Rough handling. The warhead shall withstand rough handling when tested as specified in 4.5.8 and, with the exception of the five foot drop test, shall be ballistically tested per 4.5.2 for information only. The warhead shall be safe to handle and dispose of following the five foot drop test.

3.3.5 Secured cargo vibration. The warhead shall withstand exposure to secured cargo vibration as specified in 4.5.9 and shall comply with the ballistic test as specified in 4.5.2.

3.3.6 Forty foot drop. The warhead assembled to the rocket motor and packaged for shipment shall withstand a 40 foot drop as specified in 4.5.10 and be safe to dispose of afterwards.

3.3.7 Charge circuit continuity (M439 Fuze unarmed). The continuity of the charge circuit through the umbilical lead shall be acceptable as measured by the M433/M439 Fuze Continuity Test Set. The warhead must show proper fuze continuity before assembly into a complete rocket assembly.

3.3.8 Fluoroscopic/Radiographic inspection. The Flechette warhead shall be fluoroscopically or radiographically examined to verify to the extent possible the presence and positioning of all components as well as the absence of foreign material. Warheads selected for the First Article testing shall be radiographically examined with government approved equipment prior to ballistic firing. Radiographs shall be of quality which will permit proper evaluation determined by the responsible technical agency of the government.

3.3.9 Warhead performance. The warhead has a forward ejection system that will spread approximately 1170 flechettes over a given target area in an air-to-air role along with an air-to-surface role. The warhead shall interface with the MK66 rocket motor and shall be capable of ejecting the cargo over a range of 500 to 3,000 meters by initiation of the M439 Fuze.

3.4 First article. When specified in the contract or purchase order, a sample shall be subjected to first article in accordance with the technical provisions herein (see 4.3.2).

3.5 Workmanship.

3.5.1 Mechanical. The components and assemblies shall be fabricated in a workmanlike manner. All parts and assemblies shall be free from burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, porosity, warpage, burn marks, checks, blisters, excess flash, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agents. Surface coating shall completely cover base material. All required marking shall be neat and legible.

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3.5.2 Electrical. Cable assemblies and wires shall be dressed in such manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from flux, chips, dirt, grease, rust or foreign material.

3.5.3 Product marking. The identification marking of the warhead shall be in accordance with drawing 13282664. Lot numbering shall be in accordance with MIL-STD-1168.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as a part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-A-48078 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The following types of inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

Part Description	Drawing	Quantity
Base, Nose Cone Assembly	9334110	15
Base, Nose Cone	9334113	15 *
Cone, Nose Assembly	9334119	15
Cone, Nose	9334120	15 *
Bushing, Cable Sealing	9334125	10
Plug, Base Assembly	9334127	10
Warhead Case	9334133	20
Base, Plug	9334138	15
O-Ring	9334139	10
O-Ring	9334153	10
O-Ring	9334114	10
Sleeve, Split	13282647	10
Belt Assembly Flechette: 60 Grain	13282648	10
Payload Assembly Flechette: 60 Grain	13282649	10
Flechette: 60 Grain	13282650	25
Pad, Felt	13282657	10
Loaded Case Assembly, Flechette: 60 Grain	13282663	20
Plate, Top	13282658	10
Container, Expulsion	13428931	15 *
Lid, Expulsion	13428932	15
Plate, Expulsion	13428933	15 *
Expulsion Assembly	13428934	15
Support, Fuze, Aluminum	13428935	15 *
Warhead, 2.75 Inch Flechette: XM255E1	13282664-1	100

* 15 or 5/cavity whichever is greater

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings. See Table I for inspection and testing to be performed.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

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TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Warhead, 2.75 Inch Flechette: XM255E1	1 OF 5	See Below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Base, Nose Cone Assembly (Dwg. 9334110) Examination for defects	15	3.1	4.4.2.1
	Base, Nose Cone (Dwg. 9334113) Examination for defects	15(a)	3.1	4.4.2.2
	O-Ring (Dwg. 9334114) Examination for Defects	10	3.1	4.4.2.3
	Cone, Nose Assembly (Dwg. 9334119) Examination for defects	15	3.1	4.4.2.4
	Cone, Nose (Dwg. 9334120) Examination for defects	15(a)	3.1	4.4.2.5
	Bushing, Cable Sealing (Dwg. 9334125) Examination for defects	10	3.1	4.4.2.6

NOTES:
a. 15 or 5 per cavity whichever is greater.

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TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

MIL-W-71069 (AR)

PARAGRAPH	TITLE		DRAWING NUMBER
	Warhead, 2.75 Inch Flechette: XM255E1	SHEET 2 OF 5	See Below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Plug, Base Assembly (Dwg. 9334127) Examination for defects	10	3.1	4.4.2.7
	Warhead Case (Dwg. 9334133) Examination for defects	20	3.1	4.4.2.8
	Base, Plug (Dwg. 9334138) Examination for defects	15	3.1	4.4.2.9
	O-Ring (Dwg. 9334139) Examination for Defects	10	3.1	4.4.2.10
	O-Ring (Dwg. 9334153) Examination for Defects	10	3.1	4.4.2.11
	Sleeve, Split (Dwg. 13282647) Examination for defects	10	3.1	4.4.2.12

NOTES:

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TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

MIL-W-71069 (AR)

PARAGRAPH	TITLE Warhead, 2.75 Inch Flechette: XM255E1	SHEET 3 OF 5	DRAWING NUMBER See Below
			NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Belt Assembly Flechette: 60 Grain (Dwg. 13282648) Examination for defects	10	3.1	4.4.2.13
	Payload Assembly, Flechette: 60 Grain (Dwg. 13282649) Examination for defects	10	3.1	4.4.2.14
	Flechette: 60 Grain (Dwg. 13282650) Examination for defects	25	3.1	4.4.2.15
	Pad, Felt (Dwg. 13282657) Examination for defects	10	3.1	4.4.2.16
	Plate, Top (Dwg. 13282658) Examination for defects	10	3.1	4.4.2.17

NOTES:

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

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TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

MIL-W-71069 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Warhead, 2.75 Inch Flechette: XM255E1	4 OF 5	See Below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Loaded Case Assembly, Flechette: 60 Grain (Dwg. 13282663) Examination for defects	20	3.1	4.4.2.18
	Container, Expulsion (Dwg. 13498931) Examination for defects	15(a)	3.1	4.4.2.19
	Lid, Expulsion (Dwg. 13428932) Examination for defects	15	3.1	4.4.2.20
	Plate, Expulsion (Dwg. 13428933) Examination for defects	15(a)	3.1	4.4.2.21
	Expulsion Assembly (Dwg. 13428934) Examination for defects	15	3.1	4.4.2.22
	Support, Fuze, Aluminum (Dwg. 13428935) Examination for defects	15(a)	3.1	4.4.2.23

NOTES: a. 15 or 5/ cavity whichever is greater.

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TABLE I. First article inspection.

CLASSIFICATION OF CHARACTERISTICS

MIL-W-71069 (AR)

PARAGRAPH	TITLE		DRAWING NUMBER
	Warhead, 2.75 Inch Flechette: XM255E1	SHEET 5 OF 5	See Below
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Warhead, 2.75 Inch Flechette: XM255E1 (Dwg. 13282664-1)			
	Examination for defects	100	3.1	4.4.2.24
	Weight	100	3.2.1	Scale
	Radiographic	100	3.3.8	4.5.3
	Leak Test	6	3.3.2	4.5.6
	Helicopter Vibration	19	3.3.3	4.5.7
	Rough Handling Test	36	3.3.4	4.5.8
	Secured Cargo Vibration	36	3.3.5	4.5.9
	Forty Foot Drop Test	5	3.3.6	4.5.10
	Charge Circuit Continuity	100	3.3.7	4.5.4
	Ballistic Test (b)	91(a)	3.3.9	4.5.2

NOTES:

a. Sample is composed of items from rough handling (30 - to be fired for informational purpose only), secured cargo (36), helicopter vibration (19) and Non-environmental (6).
b. The first article ballistic sample shall be rejected if one (1) or more payloads fail to eject.

AMSMC Form 1570a-E, 1 Jul 89 Replaces AMSMC Form 1570a, 1 Apr 85, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of the XM255E1 Flechette Warhead shall contain the following:

a. M439 Fuzes bearing the same metal parts lot interfix number from one metal parts manufacturer.

b. Expulsion Assemblies from lots bearing the same lot interfix number from one manufacturer.

c. Flechettes from lots bearing the same lot interfix number from one manufacturer.

The quantity of warheads in each production lot shall not exceed 5,000.

4.4.2 Examination and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table II below, using the inspection levels stated in the Classification of Characteristics paragraphs.

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TABLE II. Attributes sampling inspection.

LOT SIZE			INSPECTION LEVELS					
			I	II	III	IV	V	VI
2	to	8	*	*	*	*	*	5
9	to	15	*	*	*	*	13	5
16	to	25	*	*	*	20	13	5
26	to	50	*	*	32	20	13	5
51	to	90	*	*	32	20	13	7
91	to	150	*	125	32	20	13	11
151	to	280	*	125	32	20	20	13
281	to	500	*	125	48	47	29	16
501	to	1200	*	125	73	47	34	19
1201	to	3200	1250	125	73	53	42	23
3201	to	10000	1250	192	86	68	50	29
10001	to	35000	1250	294	108	77	60	35
35001	to	150000	1250	294	123	96	74	40

Numbers under inspection levels indicate sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative quality conformance provisions. Alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc. may be used by the contractor when they provide as a minimum, the level of quality assurance required by the provisions cited herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall, describe them in a written proposal submitted to the Government for evaluation (see 6.5). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

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CLASSIFICATION OF CHARACTERISTICS

MIL-W-71069 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Base, Nose Cone Assembly	1 OF 1	9334110
			NEXT HIGHER ASSEMBLY: 13282663

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major	None defined.			
Minor				
201	Mod 2 posts sealed improperly (2 places)	Level IV	3.1.1	Visual
202	Evidence of poor workmanship	Level IV	3.5	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-W-71069 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Base, Nose Cone *	1 OF 1	9334113
			NEXT HIGHER ASSEMBLY: 9334110

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Largest diameter incorrect	Level III	3.1.1	SMI/Gage
102	O-ring groove diameter incorrect	Level III	3.1.1	SMI/Gage
103	O-ring groove surface finish incorrect	Level III	3.1.1	Visual/Comparison Gage
Minor				
201	O-ring groove width incorrect	Level IV	3.1.1	SMI/Gage
202	Length of inside flange, minimum	Level IV	3.1.1	SMI/Gage
203	Hole diameter incorrect (2 places)	Level IV	3.1.1	SMI/Gage
204	Overall length incorrect	Level IV	3.1.1	SMI/Gage
205	Evidence of poor workmanship	Level IV	3.5	Visual

NOTES:
* In lieu of sampling inspection, the inspection of 4.5.1 may be used to qualify, mold, and maintain mold cavity qualifications.

AMSMC Form 1570b-E, 1 Jul 89 Replaces AMSMC Form 1570, 1 Feb 85, which may not be used.

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MIL-W-71069 (AR)

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.3	TITLE O-Ring	SHEET 1 OF 1	DRAWING NUMBER 9334114
			NEXT HIGHER ASSEMBLY 13282663

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Inside diameter	Level III	3.1.1	SMI/Gage
102	Cross-section diameter	Level III	3.1.1	SMI/Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.5	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-W-71069 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Cone, Nose Assembly	1 OF 1	9334119
			NEXT HIGHER ASSEMBLY: 13282664-1

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Bushing missing or improperly installed	Level III	3.1.1	Visual
102	Umbilical lead pull test	Level III	3.1.1	4.5.5
Minor				
201	Length of umbilical cable after seating crimping incorrect	Level IV	3.1.1	SMI/Gage
202	Connections incorrectly soldered	Level IV	3.1.1	Visual
203	Evidence of poor workmanship	Level IV	3.5	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-W-71069 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Cone, Nose *	1 OF 1	9334120
			NEXT HIGHER ASSEMBLY: 9334119

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Length of shoulder incorrect	Level III	3.1.1	SMI/Gage
Minor				
201	Outside diameter of shoulder incorrect	Level IV	3.1.1	SMI/Gage
202	Wall thickness of shoulder incorrect	Level IV	3.1.1	SMI/Gage
203	Largest outside diameter incorrect	Level IV	3.1.1	SMI/Gage
204	Diameter of umbilical cable hole incorrect	Level IV	3.1.1	SMI/Gage
205	Evidence of poor workmanship	Level IV	3.5	Visual

NOTES:

* In lieu of sampling inspection, the inspection of 4.5.1 may be used to qualify, mold, and maintain mold cavity qualifications.

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CLASSIFICATION OF CHARACTERISTICS

MIL-W-71069 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Bushing, Cable Sealing		9334125
			NEXT HIGHER ASSEMBLY 9334119

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major	None defined.			
Minor				
201	Hole diameter incorrect	Level IV	3.1.1	SMI/Gage
202	Radius incorrect	Level IV	3.1.1	SMI/Gage
203	Thickness incorrect	Level IV	3.1.1	SMI/Gage
204	Outside diameter incorrect	Level IV	3.1.1	SMI/Gage
205	Evidence of poor workmanship	Level IV	3.5	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-W-71069 (AR)

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.7	Plug, Base Assembly	SHEET 1 OF 1	9334127
			NEXT HIGHER ASSEMBLY 13282664-1

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major	None defined.			
Minor				
201	Felt pad missing, loose or improperly applied	Level IV	3.1.1	Visual
202	Evidence of poor workmanship	Level IV	3.5	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-W-71069 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.8	Case, Warhead		9334133
			NEXT HIGHER ASSEMBLY 13282663

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Outside thread pitch diameter incorrect	Level III	3.1.1	SMI/Gage
102	Bourrelet diameter, over maximum	Level III	3.1.1	SMI/Gage
103	Wall thickness in expand area, minimum	Level III	3.1.1	SMI/Gage
104	Nose cone base hole diameter incorrect	Level III	3.1.1	SMI/Gage
105	Inside cargo diameter incorrect	Level III	3.1.1	SMI/Gage
106	Straightness incorrect	Level III	3.1.1	SMI/Gage
107	Inside thread pitch diameter incorrect	Level III	3.1.1	SMI/Gage
108	O-ring groove diameter incorrect	Level III	3.1.1	SMI/Gage
109	O-ring groove surface finish incorrect	Level III	3.1.1	SMI/Gage/Visual
110	Chamber missing or incorrect	Level III	3.1.1	SMI/Gage
111	Machine finish 125 incorrect	Level III	3.1.1	SMI/Gage/Visual
Minor				
201	Bourrelet diameter, under minimum	Level IV	3.1.1	SMI/Gage
202	Overall length incorrect	Level IV	3.1.1	SMI/Gage
203	Base to shoulder length incorrect	Level IV	3.1.1	SMI/Gage
204	Rib height incorrect	Level IV	3.1.1	SMI/Gage
205	Fuze support hole diameter incorrect	Level IV	3.1.1	SMI/Gage
206	O-ring groove width incorrect	Level IV	3.1.1	SMI/Gage
207	Front to shoulder length incorrect	Level IV	3.1.1	SMI/Gage
208	Rib width incorrect	Level IV	3.1.1	SMI/Gage
209	Evidence of poor workmanship	Level IV	3.5	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-W-71069 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Base, Plug	1 OF 1	9334138
			NEXT HIGHER ASSEMBLY: 9334127

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Thread pitch diameter incorrect	Level III	3.1.1	SMI/Gage
102	Seal diameter incorrect	Level III	3.1.1	SMI/Gage
Minor				
201	Spanner hole diameter incorrect	Level IV	3.1.1	SMI/Gage
202	Distance between spanner holes incorrect	Level IV	3.1.1	SMI/Gage
203	Evidence of poor workmanship	Level IV	3.5	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-W-71069 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	O-Ring	1 OF 1	9334139

NEXT HIGHER ASSEMBLY: 13282663

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Inside diameter incorrect	Level III	3.1.1	SMI/Gage
102	Cross-section diameter incorrect	Level III	3.1.1	SMI/Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.5	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-W-71069 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	O-Ring	1 OF 1	9334153
			NEXT HIGHER ASSEMBLY: 13282663

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Inside diameter incorrect	Level III	3.1.1	SMI/Gage
102	Cross-section diameter incorrect	Level III	3.1.1	SMI/Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.5	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-W-71069 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.12	Sleeve, Split	1 OF 1	13282647

NEXT HIGHER ASSEMBLY: 13282649

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major	None defined.			
Minor				
201	Length incorrect	Level IV	3.1.1	SMI/Gage
202	Radius incorrect	Level IV	3.1.1	SMI/Gage
203	Thickness incorrect	Level IV	3.1.1	SMI/Gage
204	Evidence of poor workmanship	Level IV	3.5	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-W-71069 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.13	Belt Assembly, Flechette: 60 Grain	1 OF 1	13282648
			NEXT HIGHER ASSEMBLY: 13282649

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major	None defined.			
Minor				
201	Developed length incorrect	Level IV	3.1.1	SMI/Gage
202	Sewing integrity	Level IV	3.1.1	SMI/Gage
203	Evidence of poor workmanship	Level IV	3.5	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-W-71069 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.14	Payload Assembly, Flechette: 60 Grain	1 OF 1	13282649
			NEXT HIGHER ASSEMBLY 13282663

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major	None defined.			
Minor				
201	Evidence of poor workmanship	Level IV	3.5	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.15	Flechette, 60 Grain	1 OF 1	13282650
			NEXT HIGHER ASSEMBLY 13282648

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Length incorrect	Level III	3.1.1	SMI/Gage
102	Point length incorrect	Level III	3.1.1	SMI/Gage
103	Point flat maximum diameter	Level III	3.1.1	SMI/Gage
104	Hardness incorrect	(a)Level III	3.1.1	SMI/Gage
105	Overall fin length incorrect	(b)Level III	3.1.1	SMI/Gage
106	Width of fins incorrect (2 places)	(b)Level III	3.1.1	SMI/Gage
107	Half-width of fins incorrect(2 places)	(b)Level III	3.1.1	SMI/Gage
108	Root length of fins incorrect(2 places)	(b)Level III	3.1.1	SMI/Gage
109	Angular position of fins incorrect	(b)Level III	3.1.1	SMI/Gage
Minor				
201	Evidence of poor workmanship	Level IV	3.5	Visual

NOTES:
a. Minimum 4 samples/lot
b. Minimum 8 parts/hour from each press

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CLASSIFICATION OF CHARACTERISTICS

MIL-W-71069 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.16	Pad, Felt	1 OF 1	13282657

NEXT HIGHER ASSEMBLY: 13282663

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major	None defined.			
Minor				
201	Thickness incorrect	Level IV	3.1.1	SMI/Gage
202	Evidence of poor workmanship	Level IV	3.5	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-W-71069 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.17	Plate, Top	1 OF 1	13282658
			NEXT HIGHER ASSEMBLY 13282663

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Diameter incorrect	Level IV	3.1.1	SMI/Gage
Minor				
201	Thickness incorrect	Level IV	3.1.1	SMI/Gage
202	Evidence of poor workmanship	Level IV	3.5	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-W-71069 (AR)

PARAGRAPH 4.4.2.18	TITLE Loaded Case Assembly, Flechette: 60 Grain	SHEET 1 OF 1	DRAWING NUMBER 13282663
			NEXT HIGHER ASSEMBLY 13282664-1

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Charge circuit continuity	100%	3.1.1	Ohmmeter
102	Charge circuit resistance to case	100%	3.1.1	Megohmeter
103	Ground circuit continuity	100%	3.1.1	Ohmmeter
104	Pins missing, loose or above flush	Level III	3.1.1	Visual
Minor				
201	Evidence of poor workmanship	Level IV	3.5	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-W-71069 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1	DRAWING NUMBER
4.4.2.19	Container, Expulsion *			13428931
				NEXT HIGHER ASSEMBLY
				13428934
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major	None defined.			
Minor				
201	Largest outside diameter incorrect	Level IV	3.1.1	SMI/Gage
202	Smallest outside diameter incorrect	Level IV	3.1.1	SMI/Gage
203	Height from under side of lip to outside bottom incorrect	Level IV	3.1.1	SMI/Gage
204	Evidence of poor workmanship	Level IV	3.5	Visual

NOTES:

* In lieu of sampling inspection, the inspection of 4.5.1 may be used to qualify, mold, and maintain mold cavity qualifications.

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CLASSIFICATION OF CHARACTERISTICS

MIL-W-71069 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.20	Lid, Expulsion	1 OF 1	13428932

NEXT HIGHER ASSEMBLY: 13428934

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Thickness incorrect	Level III	3.1.1	SMI/Gage
Minor				
201	Outside diameter incorrect	Level IV	3.1.1	SMI/Gage
202	Evidence of poor workmanship	Level IV	3.5	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-W-71069 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.21	Plate, Expulsion	1 OF 1	13428933
			NEXT HIGHER ASSEMBLY: 13282663

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Outside diameter incorrect	Level III	3.1.1	SMI/Gage
102	O-ring groove diameter incorrect	Level III	3.1.1	SMI/Gage
103	O-ring groove width incorrect	Level III	3.1.1	SMI/Gage/Visual
Minor				
201	Thickness incorrect	Level IV	3.1.1	SMI/Gage
202	Evidence of poor workmanship	Level IV	3.5	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-W-71069 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.22	Expulsion Assembly	1 OF 1	13428934
			NEXT HIGHER ASSEMBLY 13282663

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Black powder weight incorrect	Level III	3.1.1	SMI/Scale
102	M-10 propellant weight incorrect	Level III	3.1.1	SMI/Scale
103	Expulsion lid not adhering properly to container	Level III	3.1.1	Visual
Minor				
201	Evidence of poor workmanship	Level IV	3.5	Visual

NOTES:

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CLASSIFICATION OF CHARACTERISTICS

MIL-W-71069 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.23	Support, Fuze, Aluminum *	1 OF 1	13428935
			NEXT HIGHER ASSEMBLY 13282663

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined.			
Major				
101	Largest outside diameter incorrect	Level III	3.1.1	SMI/Gage
102	Inside diameter of fuze cavity incorrect	Level III	3.1.1	SMI/Gage
103	Blow-through web thickness incorrect	Level III	3.1.1	SMI/Gage
Minor				
201	Outside diameter of case intrusion incorrect	Level IV	3.1.1	SMI/Gage
202	Depth of expulsion assembly hole incorrrect	Level IV	3.1.1	SMI/Gage
203	Largest diameter of expulsion assembly hole incorrect	Level IV	3.1.1	SMI/Gage
204	Smallest diameter of expulsion assembly hole incorrect	Level IV	3.1.1	SMI/Gage
205	Overall height incorrect	Level IV	3.1.1	SMI/Gage
206	Evidence of poor workmanship	Level IV	3.5	Visual

NOTES:

* In lieu of sampling inspection, the inspection of 4.5.1 may be used to qualify, mold, and maintain mold cavity qualifications.

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CLASSIFICATION OF CHARACTERISTICS

MIL-W-71069 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.24	Warhead, 2.75 Inch Flechette: XM255E1	1 OF 1	13282664-1
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Special				
1	Color or marking incorrect	100%	3.1.1	Visual
Major				
101	Nose cone not properly seated or gap excessive	Level III	3.1.1	SMI/Gage
102	Length, maximum	Level III	3.1.1	SMI/Gage
103	Umbilical lead, min.	Level III	3.1.1	SMI/Gage
104	Base plug protrudes beyond case base	Level III	3.1.1	Visual
105	Base O-ring missing or improperly installed	Level III	3.1.1	Visual
106	Pins missing, loose or above flush	Level III	3.1.1	Visual
107	Flouroscope	100%	3.3.8	4.5.3
108	Charge circuit continuity	100%	3.3.7	4.5.4
109	Leak test	6, 0-1	3.3.2	4.5.6
110	Ballistic functioning	36, 1-2	3.3.9	4.5.2
Minor				
201	Evidence of poor workmanship	Level IV	3.5	Visual

NOTES:

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4.4.3 Inspection equipment. The inspection equipment required to perform the inspections specified herein is identified in the "Inspection Method Reference" column in of the Classification of Characteristics listings starting with 4.4.2.1. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract. Designs which provide variable measurements instead of attributes data are preferred in order to facilitate the use of statistical process control. See MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Dimensional control of molded or cast parts. In place of normal sampling and inspection associated with the Classification of Characteristics, a sample of five parts as molded from each cavity shall be fully inspected dimensionally to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected for at least the defects listed in 4.4.2.2, 4.4.2.5, 4.4.2.19 and 4.4.2.23 after continuous production of each 5000 parts or at the end of each week, whichever occurs first. Of the sample, one part shall be the last produced. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the Contractor for inspection for each type of defect detailed in the Classification of Characteristics. All cavities removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The Contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.5.2 Ballistic functioning. The requirement for the warhead is to provide delivery of flechettes into a target area from 500 to 3000 meters from the launching helicopter. Performance shall be evaluated by ground launching a sample of 36 warheads, assembled to MK66 Mod 1, MK66 Mod 2 or MK66 Mod 3 rocket motors, from any of the compatible launchers. Twelve warheads shall be conditioned at $-50^{\circ}F \pm 5^{\circ}F$, 12 warheads at $+70^{\circ}F \pm 15^{\circ}F$ and 12 warheads at $150^{\circ}F \pm 5^{\circ}F$ for a minimum of 8 hours. Any warhead left out of the conditioning cabinet/container over five minutes prior to firing shall be placed back in the cabinet/container for not less than twice the actual time it remained out. Firing and fuze charging shall be accomplished by a Government approved Ballistic Firing Test Set using fuze time settings to achieve an airburst range of 500 and 3000 meters. Time from launch to cargo ejection and velocity at cargo ejection shall be detected and recorded. If two (2) or more payloads fail to eject the lot shall be rejected.

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4.5.3 Fluoroscopic/Radiographic. The warhead shall be x-rayed or fluoroscopically viewed in two planes, 90° opposed to verify that all major components, assemblies, subassemblies are present and properly placed, and no foreign material is present. This test is considered a failure if any major components, assemblies, or subassemblies are missing or improperly placed, if any foreign material is found, or the warhead is found unsafe. A hardcopy of the flouroscope/x-ray shall be kept on file for review.

4.5.4 Charge circuit continuity. The warhead shall be tested for proper charge circuit continuity by measuring the resistance through the umbilical lead using the M433/M439 Fuze Continuity Test Set. Any warhead which is identified as defective shall be removed from the lot or sample and held for failure analysis or rework. An open circuit continuity failure may indicate that the safety and arming device is armed; any warhead with this type of defect should be handled with appropriate safety precautions.

4.5.5 Umbilical lead pull test. A force of not less than 5 lbs. shall be applied to the umbilical lead. The cable stop eyelet shall be observed for any displacement.

4.5.6 Leakage test. Six warheads shall be subjected to the leakage test per MIL-STD-810 Method 512 Procedure I. The criteria for failure shall be water entry into the cargo compartment of the warhead. To inspect the cargo compartment, unthread the base plug in the rear of the warhead and remove the nose cone assembly and the nose cone base assembly. If one or more warheads fail to meet the requirement, the test shall be considered a failure. This test is to be conducted for first article testing and until two consecutive lots have complied with the acceptance criteria specified. After satisfying this requirement, the leakage test shall be performed on every subsequent third lot only. Should the lot fail the acceptance criteria, the contractor shall be required to return to the two consecutive lot procedures above.

4.5.7 Helicopter vibration test. The helicopter vibration test shall be performed as follows on each of nineteen (19) warheads for First Article testing only.

4.5.7.1 Sweep test. Warhead with inert motor shall be loaded into a light weight launcher and subjected to a low level sweep test. The loaded launcher shall be tested in each of three orthogonal axes. The sweep level shall be 1/2 G or lower as necessary to limit the maximum response of the launcher to the Table III levels. This test shall define the dynamic resonances of the system, as mounted on a vibration shaker.

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TABLE III. SWEEP TEST

Launcher Condition	Direction of Excitation		
	Transverse (G)	Vertical (G)	Axial (G)
Full	4.3	5.8	3.1

NOTE: The sweep time shall be 1 octave per minute 5 to 100 to 5 HZ.

4.5.7.2 Dwell test. Following the sweep tests, a warhead with inert motor loaded in a launcher shall be subjected to vibration excitation in each of the three orthogonal axes. The test shall be conducted at each of four principal frequencies 11, 22, 33 and 44 HZ. The input levels as specified in Table IV for the time durations indicated. The random noise test method shall be used and the frequency range shall be limited to the range of interest.

4.5.7.3 Fluoroscopic/Radiographic inspection. Prior to performing the helicopter vibration test, the warhead shall be x-rayed or fluoroscoped in two planes, 90° opposed, to verify that all components are secure and the safe and arming device is in the unarmed position. After performing the test, the warhead shall be x-rayed or fluoroscoped. A visual examination of the x-ray or fluoroscope shall be made and compared to the x-ray or fluoroscope made prior to vibration test. The test is considered a failure if one or more of the warheads are considered not safe as a result of the helicopter vibration test. If no safety hazard is noted, the warheads shall be ballistically tested in accordance with 4.5.2. A hardcopy of the flouroscope/x-ray shall be kept on file for review.

4.5.8 Rough handling test. The rough handling test shall be performed as follows on each of thirty-six (36) warheads for First Article testing only. Each warhead shall be packed and packaged in accordance with Dwg. 13058370-13. Inert 2.75 inch rocket motors (Dwg. 233AS431) shall be used. The test warheads are to be positioned in the box with the warhead nose in one direction and a dummy load utilized in the remaining fiber container. The dummy load shall be located in the upper right positioned as viewed from the nose end. The warheads shall be equally divided and temperature conditioned to 150°F ± 5°F, 70°F ± 5°F, and -50°F ± 5°F. The conditioning time shall be 8 hours minimum and 24 hours maximum. Each box shall be dropped seven feet onto a steel plate which is solidly supported by concrete. The orientations of the boxes on striking are:

a. Eighteen (18) warheads (6 boxes--2 from each temperature condition) warhead nose down, (longitudinal axis 45° from vertical).

b. Eighteen (18) warheads (6 boxes--2 from each temperature condition) warhead base down, (longitudinal axis 45° from vertical).

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TABLE IV. Dwell Test.

2.75 INCH ROCKET VIBRATION TEST LEVELS (MINIMUM)

CONDITION	TIME PER AXIS (HOURS)	LOC	VERTICAL AXIS					TRANSVERSE AXIS					LONGITUDINAL AXIS				
GRMS			11 HZ G^2/HZ	22 HZ G^2/HZ	33 HZ G^2/HZ	44 HZ G^2/HZ	TOT GRMS	11 HZ G^2/HZ	22 HZ G^2/HZ	33 HZ G^2/HZ	44 HZ G^2/HZ	TOT GRMS	11 HZ G^2/HZ	22 HZ G^2/HZ	33 HZ G^2/HZ	44 HZ G^2/HZ	TOT
M261 19 TUBE LOADED	.8*	FWD	.017	.05	.03	.004	.38	.015	.01	.026	.0015	.29	.03	.005	.005	.005	.3
		AFT	.05	.02	.016	.05	.6	.2	.015	.008	.004	.58	.025	.002	.0003	.002	.22
M261 19 TUBE 7 ROCKET LOAD	.63	FWD	.009	.07	.02	.715	.43	.021	.022	.007	.007	.28	.002	.01	.007	.003	.2
		AFT	.03	.10	.016	.055	.49	.008	.02	.0045	.002	.22	.006	.02	.0004	.001	.2
M261 7 TUBE LOADED	.8*	FWD	.018	.07	.02	.045	.54	.008	.21	.01	.025	.63	.007	.06	.014	.016	.35
		AFT	.08	.7	.0037	.08	1.3	.0025	1.2	.015	.06	1.45	.028	.15	.008	.017	.55
M261 7 TUBE 3 ROCKET LOAD	.63	FWD	.03	.19	.2	.05	.8	.0025	.02	.29	.03	.8	.003	.014	.01	.015	.23
		AFT	.07	.18	.13	.03	.9	.0015	.42	.1	.055	.76	.018	.008	.01	.04	.3

*For the loaded condition only, 4 minutes of the .8 hours per axis shall be at a level of 2 X the total GRMS.

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No individual box shall be subjected to more than one drop. The drop equipment and stand shall be capable of providing an unimpeded free fall prior to striking the plate and permit rebounding. Prior to performing this drop test, the following is necessary.

a. Serialize each warhead.

b. X-ray or fluoroscope each round in two planes, 90° opposed. Verify all components are secure and the fuze is unarmed.

c. Before dropping, each packaged warhead and box shall be visually examined for evidence of incorrect packaging.

After performing the drop test on all warheads, each round shall be x-rayed or fluoroscoped. A visual examination shall be performed. A hardcopy of the flouroscope/x-ray shall be kept on file for review. The test shall be rejected if one or more warheads are found to be not safe as a result of the effects of the seven-foot drop test.

4.5.8.1 Loose cargo test. If no safety hazard is noted above, the warheads will be removed from the boxes and placed back into the temperature condition chamber at the previous temperature. The conditioning time shall be 8 hours minimum and 24 hours maximum. The warheads will then be placed on an approved package tester in their fiber containers for the loose cargo test. The warheads shall not be tied down in any manner inside the tester. The tester shall then be vibrated for 30 minutes at a frequency of 5.0 hertz at 1.3 g's. Following the loose cargo test, the warheads shall be subjected to visual and x-ray or fluoroscopic examination. A hardcopy of the flouroscope/x-ray shall be kept on file for review. The test shall be rejected if one or more warheads are found to be not safe as a result of the effects of the loose cargo test. If no safety hazard is noted, thirty (30) warheads shall be ballistically tested in accordance with 4.5.2 for information only.

4.5.8.2 Five foot drop test. The remaining six (6) warheads (two conditioned to each specified temperature) shall be subjected to an unpackaged 5-foot drop test onto a steel plate which is solidly supported by concrete. The drop orientations of the warheads (at each specified temperature) on striking will be:

a. One warhead nose down with longitudinal axis 45° from vertical.

b. One warhead nose down with longitudinal axis vertical.

Following the five-foot drop test, the warheads shall be X-rayed and examined. Warheads shall be safe to handle or dispose of after the test. The First Article shall be rejected if one or more warheads are found to be not safe as a result of the effects of the five-foot drop test. Warheads subjected to the five-foot drop test shall not be ballistically tested.

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4.5.9 Secured cargo vibration test. The secured cargo vibration test shall be performed as follows on each of thirty-six (36) warheads for First Article testing only. The sample warheads shall be packed and packaged in accordance with Dwg. 13058370-13. Inert 2.75 Inch Rocket Motors (Dwg. 233AS431) shall be used. The warheads shall be equally divided and temperature conditioned to 160°F ± 5°F, 70°F ± 5°F, and -50°F ± 5°F. The conditioning time shall be 8 hours minimum and 24 hours maximum. The packed shipping boxes shall be subjected to the vibration test in accordance with Para. 4.1 International Test Operations Procedure (ITOP) 1-2-601, dated 1 March 1988. Resonance testing does not apply.

Prior to performing this test the following is necessary:

a. Serialize each round.

b. X-ray or fluoroscope each round in two planes 90° apart. Verify all components are secure and the fuze is unarmed.

Following the vibration test, each round shall be x-rayed or fluoroscoped. A visual examination shall be performed. A hardcopy of the flouroscope/x-ray shall be kept on file for review. The First Article shall be rejected if one or more warheads are found to be not safe as a result of the effects of the secured cargo vibration test. If no safety hazard is noted, the warheads shall be ballistically tested in accordance with 4.5.2.

4.5.10 Forty foot drop test. Five warheads shall be subjected to the forty foot drop test as specified in MIL-STD-331, Test 103.1, Procedure 1 (substitute warhead for fuze) for first article testing only. The warheads shall be assembled to MK66 Mod 1 inert-loaded motors and packed in accordance with Dwg. 13058370-13. One test round shall be packaged with three dummy rounds in each box. Warheads are to be X-rayed and examined for conformance to the acceptance criteria. Any round found unsafe after administering this test shall cause the first article sample to be rejected.

5. PACKAGING

5.1 Levels of protection. The following levels of protection are applicable:

a. Preservation and Packaging: Level A

b. Packing: Level A

5.1.1 Preservation and packaging. Preservation and packaging of warheads shall be in accordance with Drawing 9230116.

5.1.2 Packing. Packing of warheads shall be in accordance with Drawing 9335610.

5.1.3 Marking. Shipment(s) of warheads shall be marked in accordance with Drawing 8796521 and 8796522, and shall list DOT marking: Warhead, Ammunition with Flechettes.

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5.2 Ammunition data cards. Data card information shall be as specified in MIL-STD-1167.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The warhead covered by this specification is intended for use as part of the HYDRA-70 Rocket System.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.2).

c. Requirements for submission of first article sample.

d. Applicable stock number.

e. Packaging requirements. if other than specified in Section 5.

f. Serialization requirements, if applicable.

g. Certificate of conformance for each lot or shipment of product.

6.3 Submission of inspection equipment for design approvals. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research, Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal, Missile Command (MICOM), or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.5 Submission of alternative quality conformance provisions. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation by the technical activity responsible for the preparation of this specification. Submit copies as required to: Commander, U.S. ARDEC, ATTN: SMCAR-QAF-A, Picatinny Arsenal, NJ 07806-5000.

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6.6 Subject term (key word) listing.

ATAF
FFAR
Hydra-70
M439 Fuze
MK66 Rocket Motor
Sixty Grain

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1340-A864)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-W-71069 (AR)	19 March 1992

3. DOCUMENT TITLE

WARHEAD, 2.75 INCH ROCKET, FLECHETTE: XM255E1

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial (201) 724-6675	(2) AUTOVON 880-6675
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL,NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

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